

DW_lp_fp_multifunc_DG

Low Power Floating-Point Multi-Function with Datapath Gating

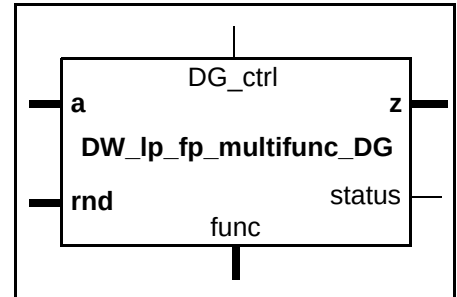
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Features and Benefits

- Has the same functionality as DW_lp_fp_multifunc when the component is in normal operation.
- Consumes less dynamic power than DW_lp_fp_multifunc when disabled. (Inputs are active, but outputs are not being used.)
- The precision format can be set for single precision or a user-defined custom floating-point format.
- Hardware for denormal numbers of IEEE 754 standard is selectively provided.
- It implements any combination of the following functions:

$$\frac{1}{x}, \frac{1}{\sqrt{x}}, \sqrt{x}, \sin(\pi x) \text{ or } \sin(x), \cos(\pi x) \text{ or } \cos(x), \log_2(x), \text{ and } 2^x$$

- One function is computed at a time.
- All functions produce monotonic results (depends on input range--see [Table 1-6](#) on page 4)
- Shared polynomial approximation unit provides a solution with reduced area.



Revision History



Note

You must use VCS to simulate the DW_lp_fp_multifunc_DG component. Non-VCS simulators are not supported.

Description

DW_lp_fp_multifunc_DG is a floating-point multi-function unit that implements any combination of seven functions: reciprocal, square root, reciprocal square root, sine, cosine, base-2 logarithm and base-2 exponential. Additionally, the component has a control input `DG_ctrl` that disables the component and reduces its dynamic power consumption when the component is not in use but the inputs are still active. The particular set of functions to be implemented is selected with the `func_select` parameter as a one-hot value. At any given time the unit computes one function in the set defined by input `func`.

The input `rnd` takes effect only when the reciprocal function is invoked with the input `func` = 1 and the parameter `faithful_round` = 0. The parameter `pi_multiple` is valid only when sine or cosine function is selected. Both input `a` and output `z` have the floating-point format, and the output `status` is an 8-bit optional status flags, which are described in the [Datapath Floating-Point Overview](#).

Table 1-1 Pin Descriptions

Pin Name	Width	Direction	Function
a	(<i>sig_width</i> + <i>exp_width</i> + 1) bits	Input	Input data
func	16 bits	Input	Function selection
DG_ctrl	1 bit	Input	Datapath gating control 0 = Component is disabled 1 = Normal component operation See “Datapath Gating Control with DG_ctrl” on page 5
rnd	3 bits	Input	Rounding mode for reciprocal function (only valid when <i>func</i> = 1)
z	(<i>sig_width</i> + <i>exp_width</i> + 1) bits	Output	Output data
status	8 bits	Output	- Status flags for the result; for details, see STATUS Flags in the <i>Datapath Floating-Point Overview</i>. <i>status</i>[7]: Indicates divide-by-zero operation of reciprocal, reciprocal square root, and logarithm functions.

Table 1-2 Parameter Description

Parameter	Values	Description
sig_width	2 to 23 Default: 23	Word length of fraction field of floating point numbers <i>a</i> and <i>z</i>
exp_width	3 to 8 Default: 8	Word length of biased exponent of floating point numbers <i>a</i> and <i>z</i>
ieee_compliance	0 or 1 Default: 0	When 1, the generated architecture includes the use of denormals and NaNs
func_select	1 to 127 Default: 127	Determines the functions to be implemented among the supported functions (one bit for each function).
faithful_round	0 or 1 Default: 1	Selects the faithful rounding mode 0 = Supports all rounding modes described in <i>Floating Point Overview</i> 1 = Results have maximum of 1 <i>ulp</i> error. Note: <i>rnd</i> is only valid when <i>func</i> = 1 (reciprocal operation)
pi_multiple	0 or 1 Default: 1	Input value (Angle) is multiplied by π This parameter is only valid when sin or cos functions are selected. 0 = $z = \sin(a)$ or $\cos(a)$ 1 = $z = \sin(\pi a)$ or $\cos(\pi a)$

Table 1-3 Synthesis Implementations

Implementation Name	Implementation	License Feature Required
rtl	Low Power Synthesis model	- DesignWare (P-2019.03 and later) - DesignWare-LP (before P-2019.03)

Table 1-4 Simulation Models

Model	Function
dw/sim_ver/DW_lp_fp_multifunc_DG.v	Verilog simulation model ^a source code Note that you must use VCS to simulate the DW_lp_fp_multifunc_DG component. Non-VCS simulators are not supported.

a. To use this simulation model, the '+v2k' options needs to be used on the VCS command line.

Parameter *func_select* and Input *func*

The 7-bit parameter *func_select* is used for selecting the set of functions to be implemented. The value of *func_select* has the range from 1 to 127, and each bit corresponds to a function, as show in [Table 1-5](#).

Table 1-5 *func_select* Parameter Implementation (design) and *func* Function Selection (use)

<i>func_select</i> Bit Weight	Function Description	<i>func</i> 16-bit select value
2^0	reciprocal, $1/a$	0000_0000_0000_0001
2^1	square root of a	0000_0000_0000_0010
2^2	reciprocal square root of a	0000_0000_0000_0100
2^3	sine, $\sin(\pi a)$	0000_0000_0000_1000
2^4	cosine, $\cos(\pi a)$	0000_0000_0001_0000
2^5	base-2 logarithm, $\log_2 a$	0000_0000_0010_0000
2^6	base-2 exponential, 2^a	0000_0000_0100_0000
	$(2^7 - 2^{15})$ reserved for future support	reserved for future support

For example, if only a reciprocal function is required, *func_select* needs to be 1 (16'h0001). If reciprocal, reciprocal square root and base-2 logarithm functions are needed, *func_select* should be 37 (16'h0025). If all seven functions are implemented, *func_select* is 127 (16'h007f).

During operation, the input port *func* specifies the single function that is to be computed at that time. The *func* port is a 16-bit input port, and it receives a one-hot encoded value. The valid one-hot codes for *func* are defined in [Table 1-5](#).

Error Range and Monotonicity

Reciprocal, square root, reciprocal square root and base-2 exponential functions admit maximum 1 *ulp* error on the normalized significand value. However, sine, cosine and base-2 logarithm functions are not applicable to this rule, but they provide the maximum $2^{-\text{sig_width}}$ error.

The DW_lp_fp_multifunc_DG component produces monotonic results. The input ranges under which monotonic results are produced are shown in [Table 1-6](#).

Table 1-6 Input Range for Monotonic Results

	1/x	sqrt(x)	1/sqrt(x)	sin(x)	cos(x)	log2(x)	exp2(x)
<i>sig_width</i> Input Range	All values of <i>sig_width</i>	All values of <i>sig_width</i>	All values of <i>sig_width</i>	<i>sig_width</i> <= 16	<i>sig_width</i> <= 16	All values of <i>sig_width</i>	All values of <i>sig_width</i>

Denormal Support

DW_lp_fp_multifunc_DG provides the hardware for denormal numbers and NaNs of IEEE 754 standard. If the parameter *ieee_compliance* is turned off, denormal numbers are considered as zeros, and NaNs are considered as infinity. Otherwise, denormal numbers and NaNs become effective and additional hardware to manipulate them is integrated.

For more information about floating point, including status flag bits, and integer and floating point formats, refer to the [Datapath Floating-Point Overview](#).

Verilog Description Example

If all functions besides sine and cosine are implemented (*func_select* = 103) and the output needs to produce the result of square root operation, the Verilog instantiation of such component is done as follows:

```
DW_lp_fp_multifunc_DG #(23, 8, 0, 103) U1 (
    .a(32'h3E800000),
    .rnd(3'b000),
    .func(16'h0002),
    .z(z),
    .status(status)
    .DG_ctrl(1'b1)
);
```

However, if the above module is used to calculate the sine function (*func* = 16'h0008), DW_lp_fp_multifunc_DG cannot generate the correct function values because it does not have the hardwired logics for the sine function with the parameter *func_select* = 103.

Datapath Gating Control with DG_ctrl

For DW_lp_fp_multifunc_DG and other combinational components that have the datapath gating feature, the DG_ctrl port is provided to control datapath gating.

When DG_ctrl = 1, the component behaves as expected according to activity on the input ports.

When DG_ctrl = 0:

- The component is disabled and internal gates are totally or partially isolated to block propagation of switching activity inside the component. This makes the component less sensitive to switching activity on the main ports and reduces dynamic power consumption.
- Values at the output ports are not defined.
- Simulation models set 'X' values at the output ports.

Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Blocks User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp.all;

entity DW_lp_fp_multifunc_DG_inst is
  generic (
    inst_sig_width : INTEGER := 23;
    inst_exp_width : INTEGER := 8;
    inst_ieee_compliance : INTEGER := 0;
    inst_func_select : INTEGER := 127;
    inst_faithful_round : INTEGER := 1;
    inst_pi_multiple : INTEGER := 1
  );
  port (
    inst_a : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_func : in std_logic_vector(15 downto 0);
    inst_rnd : in std_logic_vector(2 downto 0);
    inst_DG_ctrl : in std_logic;
    z_inst : out std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    status_inst : out std_logic_vector(7 downto 0)
  );
end DW_lp_fp_multifunc_DG_inst;

architecture inst of DW_lp_fp_multifunc_DG_inst is

begin

  -- Instance of DW_lp_fp_multifunc_DG
  U1 : DW_lp_fp_multifunc_DG
  generic map (
    sig_width => inst_sig_width,
    exp_width => inst_exp_width,
    ieee_compliance => inst_ieee_compliance,
    func_select => inst_func_select,
    faithful_round => inst_faithful_round,
    pi_multiple => inst_pi_multiple
  )
  port map (
    a => inst_a,
    func => inst_func,
    rnd => inst_rnd,
    DG_ctrl => inst_DG_ctrl,
    z => z_inst,
    status => status_inst
  );

end;
```

```

end inst;

-- pragma translate_off
configuration DW_lp_fp_multifunc_DG_inst_cfg_inst of DW_lp_fp_multifunc_DG_inst is
  for inst
    end for; -- inst
end DW_lp_fp_multifunc_DG_inst_cfg_inst;
-- pragma translate_on

```

HDL Usage Through Component Instantiation - Verilog

```

module DW_lp_fp_multifunc_DG_inst( inst_a, inst_func, inst_rnd, inst_DG_ctrl, z_inst,
status_inst );

parameter sig_width = 23;
parameter exp_width = 8;
parameter ieee_compliance = 0;
parameter func_select = 127;
parameter faithful_round = 1;
parameter pi_multiple = 1;

input [sig_width+exp_width : 0] inst_a;
input [15 : 0] inst_func;
input [2 : 0] inst_rnd;
input inst_DG_ctrl;
output [sig_width+exp_width : 0] z_inst;
output [7 : 0] status_inst;

// Instance of DW_lp_fp_multifunc_DG
DW_lp_fp_multifunc_DG #(sig_width, exp_width, ieee_compliance, func_select,
faithful_round, pi_multiple) U1 (
    .a(inst_a),
    .func(inst_func),
    .rnd(inst_rnd),
    .DG_ctrl(inst_DG_ctrl),
    .z(z_inst),
    .status(status_inst) );

endmodule

```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
January 2020	DWBB_201912.1	■ Corrected port names to be lowercase in Table 1-1 on page 2 , examples, and format presentations
March 2019	DWBB_201903.0	■ Added “ Datapath Gating Control with DG_ctrl ” on page 5 ■ Clarified some information about minPower
July 2018	DWBB_201806.1	■ For STAR 9001366625, added a note about simulator support on page 1 and in Table 1-4 on page 3 ■ Added this Revision History table and the document links on this page

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